

Simulation study: mixture of experts model with covariates

The Tien Mai & Zhi Zhao

Last updated: 01 December, 2025

Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- q covariates $\mathbf{X} = \{x_{ij}\} \in \mathbb{R}^{n \times q}$, $x_{ij} \sim N(0, 1)$
- Fixed covariate effects $\boldsymbol{\beta} \in \mathbb{R}^{q \times K}$, with $\boldsymbol{\beta}_K = 0$
- Probabilities of subpopulations $\boldsymbol{\pi} = \{\pi_{ik}\} \in \mathbb{R}^{n \times q}$, $\pi_{ik} = \exp\{\mathbf{X}_i \boldsymbol{\beta}_k\} / \sum_{l=1}^K \exp\{\mathbf{X}_i \boldsymbol{\beta}_l\}$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution $\Sigma_1, \Sigma_2, \Sigma_3$
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(moewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

# simulate data
dat <- simData(n, p,
  Xq = 3, K = 3,
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

# run a moewishart model with covariates
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishartX(S_list, K = 3, X = dat$X)
```

```
Iter    1 | LL=-2004.9 | 631.8 iter/sec
Iter  500 | LL=-1854.7 | 724.8 iter/sec
Iter 1000 | LL=-1854.0 | 735.1 iter/sec
Iter 1500 | LL=-1849.7 | 741.2 iter/sec
Iter 2000 | LL=-1856.6 | 745.7 iter/sec
Iter 2500 | LL=-1852.1 | 744.5 iter/sec
Iter 3000 | LL=-1848.8 | 744.9 iter/sec
```

```
end_time <- Sys.time()
elapsed_time <- end_time - start_time
```

```
str(fit)
```

List of 6

```
$ Beta_samples : num [1:2000, 1:3, 1:3] -0.424 -0.432 -0.427 -0.445 -0.478 ...
$ nu_samples    : num [1:2000, 1:3] 8.68 8.68 8.68 8.26 8.27 ...
$ Sigma_samples:List of 2000
..$ : num [1:2, 1:2, 1:3] 0.422 0.178 0.178 0.692 2.306 ...
..$ : num [1:2, 1:2, 1:3] 0.544 0.203 0.203 0.743 2.182 ...
..$ : num [1:2, 1:2, 1:3] 0.416 0.172 0.172 0.669 2.155 ...
..$ : num [1:2, 1:2, 1:3] 0.486 0.182 0.182 0.569 2.108 ...
..$ : num [1:2, 1:2, 1:3] 0.419 0.151 0.151 0.675 2.185 ...
..$ : num [1:2, 1:2, 1:3] 0.543 0.218 0.218 0.688 2.057 ...
..$ : num [1:2, 1:2, 1:3] 0.608 0.232 0.232 0.708 2.008 ...
..$ : num [1:2, 1:2, 1:3] 0.56 0.264 0.264 0.815 2.08 ...
..$ : num [1:2, 1:2, 1:3] 0.533 0.293 0.293 0.836 2.019 ...
..$ : num [1:2, 1:2, 1:3] 0.535 0.25 0.25 0.754 2.046 ...
..$ : num [1:2, 1:2, 1:3] 0.534 0.217 0.217 0.805 1.991 ...
..$ : num [1:2, 1:2, 1:3] 0.583 0.283 0.283 0.793 1.957 ...
..$ : num [1:2, 1:2, 1:3] 0.641 0.28 0.28 0.714 2 ...
..$ : num [1:2, 1:2, 1:3] 0.526 0.225 0.225 0.798 2.093 ...
..$ : num [1:2, 1:2, 1:3] 0.523 0.23 0.23 0.89 1.963 ...
..$ : num [1:2, 1:2, 1:3] 0.591 0.23 0.23 0.869 1.942 ...
..$ : num [1:2, 1:2, 1:3] 0.53 0.2 0.2 0.704 2.023 ...
..$ : num [1:2, 1:2, 1:3] 0.584 0.261 0.261 0.769 1.861 ...
..$ : num [1:2, 1:2, 1:3] 0.532 0.189 0.189 0.762 1.84 ...
..$ : num [1:2, 1:2, 1:3] 0.537 0.247 0.247 0.797 2.113 ...
..$ : num [1:2, 1:2, 1:3] 0.536 0.197 0.197 0.768 1.886 ...
..$ : num [1:2, 1:2, 1:3] 0.563 0.263 0.263 0.87 1.871 ...
..$ : num [1:2, 1:2, 1:3] 0.604 0.239 0.239 0.814 1.668 ...
..$ : num [1:2, 1:2, 1:3] 0.545 0.186 0.186 0.667 1.747 ...
..$ : num [1:2, 1:2, 1:3] 0.466 0.174 0.174 0.595 1.679 ...
..$ : num [1:2, 1:2, 1:3] 0.515 0.233 0.233 0.641 1.732 ...
..$ : num [1:2, 1:2, 1:3] 0.409 0.136 0.136 0.624 1.655 ...
..$ : num [1:2, 1:2, 1:3] 0.453 0.279 0.279 0.824 1.749 ...
..$ : num [1:2, 1:2, 1:3] 0.463 0.198 0.198 0.773 1.725 ...
..$ : num [1:2, 1:2, 1:3] 0.508 0.263 0.263 0.874 1.852 ...
..$ : num [1:2, 1:2, 1:3] 0.552 0.187 0.187 0.755 1.873 ...
..$ : num [1:2, 1:2, 1:3] 0.52 0.211 0.211 0.74 1.925 ...
..$ : num [1:2, 1:2, 1:3] 0.529 0.248 0.248 0.896 1.749 ...
..$ : num [1:2, 1:2, 1:3] 0.538 0.224 0.224 0.85 1.921 ...
..$ : num [1:2, 1:2, 1:3] 0.583 0.233 0.233 0.787 1.913 ...
..$ : num [1:2, 1:2, 1:3] 0.482 0.198 0.198 0.743 2.074 ...
..$ : num [1:2, 1:2, 1:3] 0.448 0.226 0.226 0.711 2.275 ...
..$ : num [1:2, 1:2, 1:3] 0.475 0.229 0.229 0.682 2.231 ...
..$ : num [1:2, 1:2, 1:3] 0.448 0.184 0.184 0.642 2.41 ...
..$ : num [1:2, 1:2, 1:3] 0.469 0.177 0.177 0.527 1.964 ...
..$ : num [1:2, 1:2, 1:3] 0.402 0.194 0.194 0.619 2.367 ...
..$ : num [1:2, 1:2, 1:3] 0.383 0.129 0.129 0.616 2.15 ...
..$ : num [1:2, 1:2, 1:3] 0.367 0.108 0.108 0.662 2.194 ...
..$ : num [1:2, 1:2, 1:3] 0.348 0.114 0.114 0.596 1.957 ...
..$ : num [1:2, 1:2, 1:3] 0.427 0.184 0.184 0.651 2.004 ...
```

```

..$ : num [1:2, 1:2, 1:3] 0.453 0.171 0.171 0.576 1.931 ...
..$ : num [1:2, 1:2, 1:3] 0.474 0.247 0.247 0.577 2.015 ...
..$ : num [1:2, 1:2, 1:3] 0.446 0.218 0.218 0.633 1.987 ...
..$ : num [1:2, 1:2, 1:3] 0.442 0.23 0.23 0.777 2.011 ...
..$ : num [1:2, 1:2, 1:3] 0.395 0.14 0.14 0.637 1.918 ...
..$ : num [1:2, 1:2, 1:3] 0.422 0.154 0.154 0.54 2.155 ...
..$ : num [1:2, 1:2, 1:3] 0.352 0.128 0.128 0.579 1.967 ...
..$ : num [1:2, 1:2, 1:3] 0.369 0.116 0.116 0.54 1.943 ...
..$ : num [1:2, 1:2, 1:3] 0.38 0.116 0.116 0.507 1.846 ...
..$ : num [1:2, 1:2, 1:3] 0.354 0.156 0.156 0.525 1.92 ...
..$ : num [1:2, 1:2, 1:3] 0.341 0.135 0.135 0.47 1.99 ...
..$ : num [1:2, 1:2, 1:3] 0.36 0.166 0.166 0.472 1.926 ...
..$ : num [1:2, 1:2, 1:3] 0.35 0.152 0.152 0.545 2.001 ...
..$ : num [1:2, 1:2, 1:3] 0.361 0.152 0.152 0.504 2.136 ...
..$ : num [1:2, 1:2, 1:3] 0.365 0.137 0.137 0.496 1.978 ...
..$ : num [1:2, 1:2, 1:3] 0.34 0.157 0.157 0.519 1.923 ...
..$ : num [1:2, 1:2, 1:3] 0.366 0.138 0.138 0.561 2.11 ...
..$ : num [1:2, 1:2, 1:3] 0.371 0.129 0.129 0.52 2.038 ...
..$ : num [1:2, 1:2, 1:3] 0.357 0.132 0.132 0.499 2.178 ...
..$ : num [1:2, 1:2, 1:3] 0.425 0.187 0.187 0.548 2 ...
..$ : num [1:2, 1:2, 1:3] 0.381 0.178 0.178 0.523 1.97 ...
..$ : num [1:2, 1:2, 1:3] 0.434 0.208 0.208 0.555 1.774 ...
..$ : num [1:2, 1:2, 1:3] 0.422 0.164 0.164 0.55 2.042 ...
..$ : num [1:2, 1:2, 1:3] 0.393 0.158 0.158 0.541 2.094 ...
..$ : num [1:2, 1:2, 1:3] 0.373 0.155 0.155 0.563 2.127 ...
..$ : num [1:2, 1:2, 1:3] 0.359 0.144 0.144 0.493 1.921 ...
..$ : num [1:2, 1:2, 1:3] 0.39 0.173 0.173 0.582 1.862 ...
..$ : num [1:2, 1:2, 1:3] 0.376 0.144 0.144 0.505 1.949 ...
..$ : num [1:2, 1:2, 1:3] 0.413 0.184 0.184 0.556 2.008 ...
..$ : num [1:2, 1:2, 1:3] 0.368 0.195 0.195 0.644 1.92 ...
..$ : num [1:2, 1:2, 1:3] 0.431 0.184 0.184 0.599 1.842 ...
..$ : num [1:2, 1:2, 1:3] 0.441 0.168 0.168 0.581 1.893 ...
..$ : num [1:2, 1:2, 1:3] 0.423 0.167 0.167 0.609 1.794 ...
..$ : num [1:2, 1:2, 1:3] 0.4 0.161 0.161 0.704 1.982 ...
..$ : num [1:2, 1:2, 1:3] 0.377 0.137 0.137 0.605 1.82 ...
..$ : num [1:2, 1:2, 1:3] 0.501 0.162 0.162 0.598 1.774 ...
..$ : num [1:2, 1:2, 1:3] 0.444 0.14 0.14 0.614 1.681 ...
..$ : num [1:2, 1:2, 1:3] 0.498 0.217 0.217 0.666 1.617 ...
..$ : num [1:2, 1:2, 1:3] 0.49 0.241 0.241 0.669 1.726 ...
..$ : num [1:2, 1:2, 1:3] 0.427 0.124 0.124 0.583 1.818 ...
..$ : num [1:2, 1:2, 1:3] 0.444 0.146 0.146 0.575 1.778 ...
..$ : num [1:2, 1:2, 1:3] 0.426 0.202 0.202 0.669 1.745 ...
..$ : num [1:2, 1:2, 1:3] 0.399 0.189 0.189 0.684 1.848 ...
..$ : num [1:2, 1:2, 1:3] 0.405 0.188 0.188 0.669 1.911 ...
..$ : num [1:2, 1:2, 1:3] 0.384 0.132 0.132 0.63 1.919 ...
..$ : num [1:2, 1:2, 1:3] 0.416 0.15 0.15 0.618 1.701 ...
..$ : num [1:2, 1:2, 1:3] 0.458 0.143 0.143 0.575 1.594 ...
..$ : num [1:2, 1:2, 1:3] 0.459 0.205 0.205 0.696 1.671 ...
..$ : num [1:2, 1:2, 1:3] 0.486 0.142 0.142 0.624 1.799 ...
..$ : num [1:2, 1:2, 1:3] 0.502 0.16 0.16 0.65 1.882 ...
..$ : num [1:2, 1:2, 1:3] 0.542 0.197 0.197 0.723 1.861 ...
..$ : num [1:2, 1:2, 1:3] 0.569 0.291 0.291 0.803 1.928 ...
..$ : num [1:2, 1:2, 1:3] 0.51 0.231 0.231 0.796 1.847 ...
..$ : num [1:2, 1:2, 1:3] 0.545 0.201 0.201 0.739 1.92 ...

```

```

.. [list output truncated]
$ z_samples      : int [1:2000, 1:200] 1 1 1 1 1 1 1 1 1 1 ...
$ pi_mean        : num [1:200, 1:3] 0.51227 0.57043 0.29485 0.00712 0.63901 ...
$ loglik         : num [1:3000] -2005 -1998 -1988 -1980 -1977 ...

# posterior summaries

colMeans(fit$pi) # estimated pis

[1] 0.2579648 0.4227126 0.3193227

colMeans(dat$pi) # true pis

[1] 0.2935523 0.4092209 0.2972268

apply(fit$nu, 2, mean) # estimated nu

[1] 8.063060 13.807920 3.383995

dat$nu # true nu

[1] 8 12 3

# Sum all arrays in the list and divide by the number of saved iterations
MoE_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(MoE_mean[, , 1])

      [,1]      [,2]
[1,] 0.5322461 0.2195507
[2,] 0.2195507 0.7717931

print(MoE_mean[, , 2])

      [,1]      [,2]
[1,] 1.807168 0.569296
[2,] 0.569296 1.361460

print(MoE_mean[, , 3])

      [,1]      [,2]
[1,] 4.0278828 -0.1476875
[2,] -0.1476875 2.9776286

as.data.frame(Sigma_list)

   X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2 2.0 0.6 4.0 0.2
2 0.2 0.7 0.6 1.5 0.2 3.0

(a2 <- apply(MoE_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa - x)^2)))))

[1] 0.006958526 0.058263106 0.243051109

```

EM algorithm

```

# run the model with EM algorithm
set.seed(1234)
fit3 <- mewishartX(S_list, K = 3, X = dat$X, method = "em")

Iter    1  loglik = -2083.438934

```

```
Iter 2 loglik = -2080.061603
Iter 3 loglik = -2074.927308
Iter 4 loglik = -2052.031223
Iter 5 loglik = -2009.809851
Iter 6 loglik = -1979.977069
Iter 7 loglik = -1963.808984
Iter 8 loglik = -1944.596201
Iter 9 loglik = -1920.714908
Iter 10 loglik = -1909.106972
Iter 11 loglik = -1906.747217
Iter 12 loglik = -1905.919572
Iter 13 loglik = -1905.306686
Iter 14 loglik = -1904.987661
Iter 15 loglik = -1904.877908
Iter 16 loglik = -1904.846361
Iter 17 loglik = -1904.837700
Iter 18 loglik = -1904.835340
Iter 19 loglik = -1904.834694
Iter 20 loglik = -1904.834521
Iter 21 loglik = -1904.834472
Iter 22 loglik = -1904.834458
Iter 23 loglik = -1904.834453
Iter 24 loglik = -1904.834451
Iter 25 loglik = -1904.834451
Converged by loglik tolerance.
```

```
# EM estimates
colMeans(fit3$gamma)
```

```
[1] 0.4515924 0.2544287 0.2939790
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.2935523 0.4092209 0.2972268
```

```
fit3$nu
```

```
[1] 7 7 7
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```